

Section : Chemistry

Q.1 Arrange the following electrolytes in the increasing order of van't Hoff factor (i) for complete dissociation of solute.

- (A) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$
 (B) MgCl_2
 (C) $\text{K}_4[\text{Fe}(\text{CN})_6]$
 (D) KCl

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (D), (B), (A), (C)
3. (D), (A), (B), (C)
4. (C), (A), (B), (D)

Options 1. 1

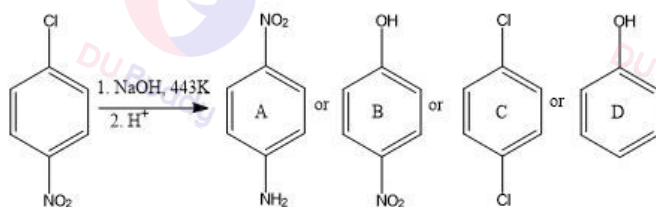
2. 2

3. 3

4. 4

Q.2

The product in the following reaction is



1. A
2. B
3. C
4. D

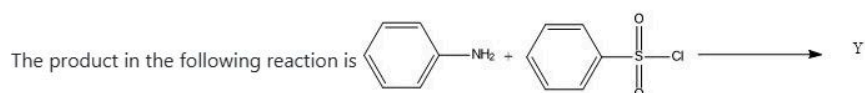
Options 1. 1

2. 2

3. 3

4. 4

Q.3



- 1.
- 2.
- 3.
- 4.

Options 1. 1

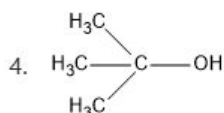
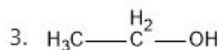
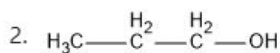
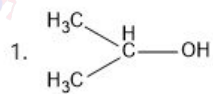
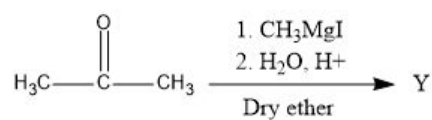
2. 2

3. 3

4. 4

Q.4

The product Y in the following reactions



Options 1. 1

2. 2

3. 3

4. 4

Q.5 The amines which are soluble in water

- (A) Methanamine
(B) Ethanamine
(C) Propanamine
(D) Benzylamine

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (B), (C) and (D) only
3. (A), (B), (C) and (D)
4. (A), (B) and (C) only

Options 1. 1

2. 2

3. 3

4. 4

Q.6 Arrange the following in increasing order of their pK_a values.

- (A) o-nitrophenol
(B) m-nitrophenol
(C) Phenol
(D) Ethanol

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (B), (A), (C), (D)
3. (D), (C), (B), (A)
4. (B), (A), (C), (D)

Options 1. 1

2. 2

3. 3

4. 4

Q.7 Interstitial compounds are formed when small atoms like H, C, or N are trapped in the crystal lattice of metal. Which of the following statements are true about the interstitial compounds?

- (A) Interstitial compounds have higher melting points than the pure metals.
(B) Interstitial compounds are very soft.
(C) Interstitial compounds are very reactive.
(D) Interstitial compounds retain metallic conductivity.

Choose the correct answer from the options given below:

1. (A), and (D) only

2. (A), (B) and (D) only
3. (B), (C) and (D) only
4. (B) and (C) only

Options 1. 1
2. 2
3. 3
4. 4

Q.8 Match List-I (Common name) with List-II (IUPAC name)

List-I	List-II
Common Name	IUPAC name
(A) Aniline	(I) N-Phenylethanamide
(B) Acetanilide	(II) Benzenamine
(C) Sulphanilic acid	(III) Prop-2-en-1-amine
(D) Allylamine	(IV) p-aminobenzene sulphonic acid

Choose the correct answer from the options given below:

1. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)
2. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1
2. 2
3. 3
4. 4

Q.9 Magnetic moment values for $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ and $[\text{Fe}(\text{CN})_6]^{3-}$ are

1. 1.74 BM and 5.92 respectively
2. 5.92 BM and 1.74 respectively
3. 1.74 BM for both
4. 5.92 BM for both

Options 1. 1
2. 2
3. 3
4. 4

Q.10 Arrange the following electrodes in decreasing order of the strength of the respective ions as reducing agent.

- (A) $\text{Cu}^{2+} + 2e^- \rightarrow \text{Cu}(s) \quad E^\circ = 0.34 \text{ V}$
- (B) $\text{Sn}^{2+} + 2e^- \rightarrow \text{Sn}(s) \quad E^\circ = -0.14 \text{ V}$
- (C) $\text{AgBr}(s) + e^- \rightarrow \text{Ag}(s) + \text{Br}^- \quad E^\circ = 0.10 \text{ V}$
- (D) $\text{Pb}^{2+} + 2e^- \rightarrow \text{Pb}(s) \quad E^\circ = -0.13 \text{ V}$

Choose the correct answer from the options given below:

1. (B), (D), (C), (A)
2. (A), (C), (D), (B)
3. (C), (A), (B), (D)
4. (D), (B), (A), (C)

Options 1. 1
2. 2
3. 3
4. 4

Q.11 The mass of NaOH required to make a 500 mL solution of 0.5 molarity is

1. 15 g
2. 20 g
3. 40 g
4. 10 g

Options 1. 1
2. 2
3. 3
4. 4

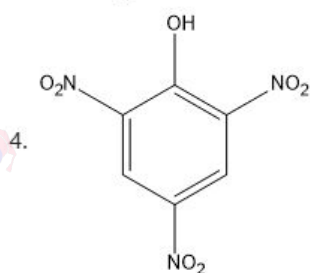
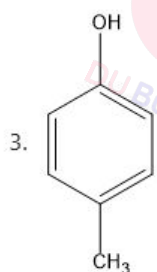
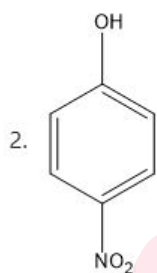
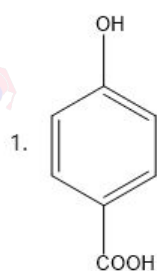
Q.12 Identify the carbohydrate which is not a polysaccharide

1. Starch
2. Cellulose
3. Canesugar
4. Glycogen

Options 1. 1
2. 2
3. 3

Q.13

The structure of picric acid is



Options 1. 1

2. 2

3. 3

4. 4

Q.14 Osmotic pressure of 0.02 M urea solution at 27°C, is (given $R = 0.0821 \text{ atm L mol}^{-1} \text{ K}^{-1}$)

1. 0.4926 atm

2. 0.8321 atm

3. 1.50 atm

4. 2.00 atm

Options 1. 1

2. 2

3. 3

4. 4

Q.15 In the electrochemical series written in the increasing order of reduction potential
 (A) The elements which are placed above the hydrogen in the electrochemical series are stronger oxidizing agents.
 (B) The elements which are placed above the hydrogen in the electrochemical series are stronger reducing agents.
 (C) The elements with the positive electrode potential values lie below the hydrogen in the electrochemical series.
 (D) The activity of both the metals and non-metals in the electrochemical series decreases from top to bottom.

Choose the correct answer from the options given below:

1. (B) and (C) only

2. (A), (B) and (D) only

3. (A), (B), (C) only
4. (B), (C) and (D) only

Options 1. 1
2. 2
3. 3
4. 4

Q.16 Match the ions given in List-I with number of unpaired electrons given in List-II

List-I	List-II
Ion	Number of unpaired electrons
(A) V^{2+}	(I) 1
(B) Cr^{2+}	(II) 4
(C) Ti^{2+}	(III) 3
(D) Cu^{2+}	(IV) 2

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
2. (A) - (III), (B) - (II), (C) - (I), (D) - (IV)
3. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
4. (A) - (III), (B) - (II), (C) - (IV), (D) - (I)

Options 1. 1
2. 2
3. 3
4. 4

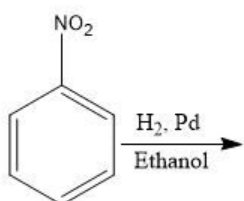
Q.17 The correct cell representation for the following reaction is



1. $Mg(s)|Mg^{2+}(0.130M)||Ag^+(0.0001M)|Ag(s)$
2. $Mg(s)||Mg^{2+}(0.130M)|Ag^+(0.001M)||Ag(s)$
3. $Mg(s)|Mg^{2+}(0.130M)||Ag(s)|Ag^+(0.0001M)$
4. $Mg^{2+}(0.130M)|Mg(s)||Ag(s)|Ag^+(0.0001M)$

Options 1. 1
2. 2
3. 3
4. 4

Q.18 Reduction of nitrobenzene gives



1. Nitrotoluene
2. Aniline
3. Benzoic acid
4. Benzene

Options 1. 1
2. 2
3. 3
4. 4

Q.19 Magnetic moment of first series of transition metal complexes is determined by.....

1. subtracting spin angular momentum from orbital angular momentum
2. Orbital angular momentum only
3. spin angular momentum only
4. product of spin angular momentum and orbital angular momentum

Options 1. 1
2. 2
3. 3
4. 4

Q.20 From the statements given below, choose the correct statement regarding the galvanic cell.

1. Positively charged electrode is called anode.
2. Negatively charged electrode is called cathode.
3. Oxidation occurs at cathode.
4. Reduction occurs at cathode.

Options 1. 1
2. 2
3. 3
4. 4

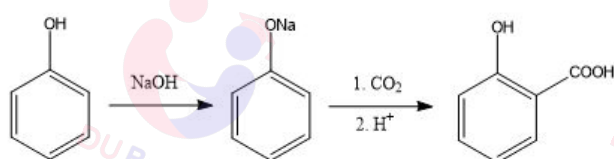
Q.21 The specific conductance of a decinormal KCl solution is $0.0121 \text{ ohm}^{-1} \text{ cm}^{-1}$ and the resistance was found to be 56 ohm. Cell constant for the reaction will be

1. 0.6776 cm^{-1}
2. 1.212 cm^{-1}
3. 1.667 cm^{-1}
4. 1.121 cm^{-1}

Options 1. 1
2. 2
3. 3
4. 4

Q.22

The reaction for the conversion of phenol into salicylic acid is known as



1. Reimer-Tiemann reaction
2. Kolbe's reaction
3. Claisen-condensation reaction
4. Etard reaction

Options 1. 1
2. 2
3. 3
4. 4

Q.23 Match List-I (Amino acid) with List-II (Example)

List-I	List-II
Amino acid	Example
(A) Basic amino acid	(I) Glycine
(B) Acidic amino acid	(II) Alanine
(C) Neutral amino acid	(III) Aspartic acid
(D) Optically inactive amino acid	(IV) Lysine

Choose the correct answer from the options given below:

1. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
2. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
3. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1
2. 2
3. 3
4. 4

Q.24 Which of the following complexes will show absorbance of light at the lowest wavelength?

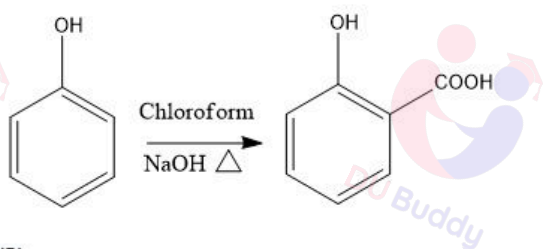
1. $[\text{Co}(\text{NH}_3)_6]^{3+}$
2. $[\text{CoCl}(\text{NH}_3)_5]^{2+}$
3. $[\text{Co}(\text{NH}_3)_5(\text{H}_2\text{O})]^{3+}$
4. $[\text{Co}(\text{CN})_6]^{3-}$

Options 1. 1
2. 2
3. 3
4. 4

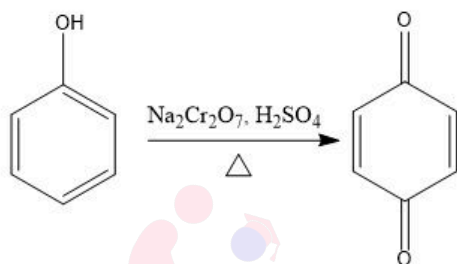
Q.25

The conversions which are represented correctly are

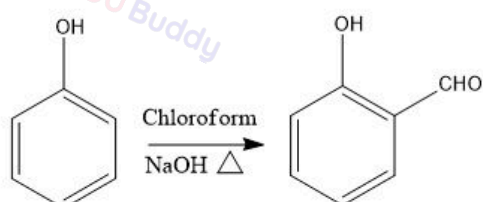
(A)



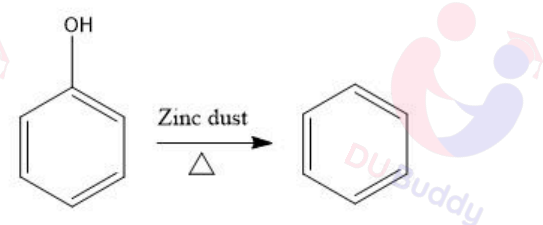
(B)



(C)



(D)

Choose the **correct** answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.26 Match List-I with List-II

List-I

Term

(A) Raoult's law

(B) Henry's Law

(C) Dalton's Law of partial pressure

(D) Relative lowering of vapor pressure (IV)

List-II

Formula/expression

(I)

$$\frac{p_1^\circ - p_1}{p_1^\circ} = x_2$$

(II)

$$p_{\text{total}} = p_1^\circ + (p_2^\circ - p_1^\circ)x_2$$

(III)

$$p = K_H x$$

List-I

List-II

$$p_2 = p_2^2 x_2$$

Choose the correct answer from the options given below:

1. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
2. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
3. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)
4. (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

Options 1. 1

2. 2

3. 3

4. 4

Q.27 V_2O_5 catalyst is used in the manufacturing of

1. Ammonia
2. Urea
3. Sulphuric acid
4. Nitric acid

Options 1. 1

2. 2

3. 3

4. 4

Q.28 The coordination number (CN) and oxidation state (OS) of Co in $[Co(C_2O_4)_3]^{3-}$ is.....

1. CN 3 and OS +3 respectively
2. CN 6 and OS +3 respectively
3. CN 6 and OS +6 respectively
4. CN 3 and OS +6 respectively

Options 1. 1

2. 2

3. 3

4. 4

Q.29 Haloalkanes can be represented by the formula

1. $C_nH_{2n+2}X$
2. $C_nH_{2n+1}X$
3. $C_nH_{2n}X$
4. $C_nH_{n+2}X$

Options 1. 1

2. 2

3. 3

4. 4

Q.30 n-Hexane is obtained when glucose is

1. Prolonged heated with HI
2. Reacted with hydroxylamine.
3. Treated with mild oxidising agent like bromine water.
4. Treated with acetic anhydride.

Options 1. 1

2. 2

3. 3

4. 4

Q.31 Which of the following is correct about the nature of ligand?

1. CN^- is an ambidentate ligand
2. Ethylenediaminetetraacetate ion ($EDTA^{4-}$) is tetradentate ligand
3. CH_3NH_2 is a bidentate ligand
4. $C_2O_4^{2-}$ is not a Chelating ligand

Options 1. 1

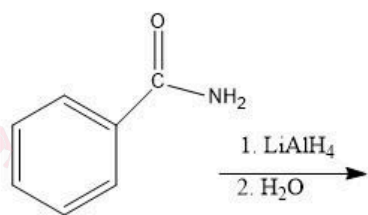
2. 2

3. 3

4. 4

Q.32

The product in the following reaction is



- 1.
- 2.
- 3.
- 4.

Options

1. 1
2. 2
3. 3
4. 4

Q.33 Identify

Identify the optically active compounds from the following:

- (A) $[\text{Co}(\text{en})_3]^{3+}$
- (B) $\text{trans}-[\text{Co}(\text{en})_2\text{Cl}_2]^+$
- (C) $\text{cis}-[\text{Co}(\text{en})_2\text{Cl}_2]^+$
- (D) $[\text{Cr}(\text{NH}_3)_5\text{Cl}]$

Choose the correct answer from the options given below:

1. (A), (C) and (D) only
2. (A), (B) and (D) only
3. (A), and (C) only
4. (B) and (D) only

Options

1. 1
2. 2
3. 3
4. 4

Q.34 Arrange the following elements in decreasing order of their radii:

- (A) La
- (B) Pr
- (C) Gd
- (D) Yb

Choose the correct answer from the options given below:

1. (B), (A), (C), (D)
2. (A), (B), (C), (D)
3. (D), (A), (C), (B)

4. (A), (B), (D), (C)

Options 1. 1

2. 2

3. 3

4. 4

Q.35 Match List-I with List-II

List-I (Name)

List-II (Use)

(A) Carbon tetrachloride (I) Pesticide

(B) DDT (II) Solvent

(C) Iodoform (III) Refrigerant

(D) Freons-12 (IV) Antiseptic

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)

2. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)

3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)

4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

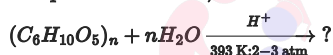
Options 1. 1

2. 2

3. 3

4. 4

Q.36 The product obtained, when starch is hydrolyzed with dilute sulphuric acid at 393 K under pressure, is



Options:

1. Glucose only

2. A mixture of glucose and fructose

3. Fructose only

4. Gluconic acid

Options 1. 1

2. 2

3. 3

4. 4

Q.37 Which statement is correct for solutions?

1. Ideal solutions do not obey Raoult's law.

2. For an ideal solution, $\Delta H_{\text{mix}} > 0$ and $\Delta V_{\text{mix}} > 0$

3. If $\Delta H_{\text{mix}} > 0$ and $\Delta V_{\text{mix}} > 0$, then solutions show negative deviations.

4. If $\Delta H_{\text{mix}} > 0$ and $\Delta V_{\text{mix}} > 0$, then solutions show positive deviations.

Options 1. 1

2. 2

3. 3

4. 4

Q.38 Arrange the following in order of increasing boiling points

(A) 1,4-Dichlorobenzene

(B) 1,3-Dichlorobenzene

(C) 1,2-Dichlorobenzene

(D) Chlorobenzene

Choose the correct answer from the options given below:

1. (D), (C), (B), (A)

2. (A), (B), (C), (D)

3. (B), (A), (D), (C)

4. (D), (B), (A), (C)

Options 1. 1

2. 2

3. 3

4. 4

Q.39 Hybridization of carbon atoms attached with halogens in haloalkanes & haloarenes are

1. sp^2 & sp^2 hybridized respectively

2. sp^3 & sp^2 hybridized respectively

3. sp^3 & sp^3 hybridized respectively

4. sp^2 & sp^3 hybridized respectively

Options 1. 1

2. 2

3. 3

Q.40 Type of RNA which does not exist is

1. m-RNA
2. s-RNA
3. r-RNA
4. t-RNA

Options 1. 1

2. 2
3. 3
4. 4

Q.41 Read the Passage carefully and answer the Questions

Kinetic studies not only help us to determine the rate of a chemical reaction but also describe the conditions by which the reaction rates can be altered. The factors such as concentration, temperature, pressure and catalyst affect the rate of a reaction.

The speed of a reaction or the rate of a reaction can be defined as the change in concentration of a reactant or product in unit time.

$$\text{Rate} = k[A]^x[B]^y$$

Here k is the rate constant

$x + y$ gives the overall order of a reaction, whereas x and y represent the order with respect to the reactants A and B , respectively.

Which of the following statements are correct?

- (A) Catalysts increase the rate of a chemical reaction.
- (B) Catalysts can catalyze both spontaneous and non-spontaneous reactions.
- (C) Higher the activation energy for a chemical reaction, lower is the rate of the reaction.
- (D) Catalyst increases the potential energy barrier.

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A) and (C) only
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.42 Read the Passage carefully and answer the Questions

Kinetic studies not only help us to determine the rate of a chemical reaction but also describe the conditions by which the reaction rates can be altered. The factors such as concentration, temperature, pressure and catalyst affect the rate of a reaction.

The speed of a reaction or the rate of a reaction can be defined as the change in concentration of a reactant or product in unit time.

$$\text{Rate} = k[A]^x[B]^y$$

Here k is the rate constant

$x + y$ gives the overall order of a reaction, whereas x and y represent the order with respect to the reactants A and B , respectively.

Table: Initial rate of formation of NO₂

S.No. Initial [NO]/mol L⁻¹ Initial [O₂]/mol L⁻¹ Initial rate of formation of NO₂/mol L⁻¹s⁻¹

1	0.30	0.30	0.096
2	0.60	0.30	0.384
3	0.30	0.60	0.192
4	0.60	0.60	0.768

Based on the above observations, the rate of formation of NO₂ can be represented as

$$1. \text{Rate of reaction} = k[\text{NO}]^2[\text{O}_2]$$

$$2. \text{Rate of reaction} = k[\text{NO}][\text{O}_2]^2$$

$$3. \text{Rate of reaction} = k[\text{NO}]^2[\text{O}_2]^2$$

$$4. \text{Rate of reaction} = k[\text{NO}][\text{O}_2]$$

Options 1. 1

2. 2
3. 3
4. 4

Q.43 Read the Passage carefully and answer the Questions

Kinetic studies not only help us to determine the rate of a chemical reaction but also describe the conditions by which the reaction rates can be altered. The factors such as concentration, temperature, pressure and catalyst affect the rate of a reaction.

The speed of a reaction or the rate of a reaction can be defined as the change in concentration of a reactant or product in unit time.

$$\text{Rate} = k[A]^x[B]^y$$

Here k is the rate constant

$x + y$ gives the overall order of a reaction, whereas x and y represent the order with respect to the reactants A and B , respectively.

The order of a reaction which does not exist is

1. Zero order reaction
2. First order reaction
3. Pseudo first order reaction
4. Pseudo second order reaction

Options 1. 1

2. 2

3. 3

4. 4

Q.44 Read the Passage carefully and answer the Questions

Kinetic studies not only help us to determine the rate of a chemical reaction but also describe the conditions by which the reaction rates can be altered.

The factors such as concentration, temperature, pressure and catalyst affect the rate of a reaction.

The speed of a reaction or the rate of a reaction can be defined as the change in concentration of a reactant or product in unit time.

$$\text{Rate} = k[A]^x[B]^y$$

Here k is the rate constant

$x + y$ gives the overall order of a reaction, whereas x and y represent the order with respect to the reactants A and B , respectively.

The unit of rate constant for first order reaction is

1. $\text{mol L}^{-1} \text{s}^{-1}$
2. mol L s^{-1}
3. $\text{L}^{-1} \text{s}^{-1}$
4. s^{-1}

Options 1. 1

2. 2

3. 3

4. 4

Q.45 Read the Passage carefully and answer the Questions

Kinetic studies not only help us to determine the rate of a chemical reaction but also describe the conditions by which the reaction rates can be altered.

The factors such as concentration, temperature, pressure and catalyst affect the rate of a reaction.

The speed of a reaction or the rate of a reaction can be defined as the change in concentration of a reactant or product in unit time.

$$\text{Rate} = k[A]^x[B]^y$$

Here k is the rate constant

$x + y$ gives the overall order of a reaction, whereas x and y represent the order with respect to the reactants A and B , respectively.

If the temperature is raised by 10°C , then the rate of the reaction increases approximately by

1. Two times
2. Three times
3. Five times
4. Ten times

Options 1. 1

2. 2

3. 3

4. 4

Q.46 Read the passage carefully and answer the Questions

Carbonyl compounds such as aldehydes, ketones, and carboxylic acids play an important role in the biochemical process of plants and animal kingdom.

Many aldehydes and ketones find applications in dye, perfumery and other industries.

The carbonyl carbon atom is sp^2 -hybridized forming three sigma and one pi-bond with oxygen.

C-O bonds are polar bond due to high electronegativity of oxygen than carbon.

Hence the carbonyl carbon behaves as an electrophilic center and oxygen as a nucleophilic centre.

Primary and secondary alcohols yield aldehydes and ketones respectively upon oxidation with variety of oxidizing agents.

The boiling points of aldehydes and ketones differ from hydrocarbons and ethers.

Aldehydes and ketones undergo nucleophilic addition reaction due to presence of electrophilic carbonyl carbon.

Nucleophile and a proton add across the carbon oxygen double bond.

The starting compound Y in the following reaction is



1. RCH_2COOH
2. $\text{RCH} = \text{CHOH}$
3. RCH_2OH
4. RCH_2CH_3

Options 1. 1

2. 2
3. 3
4. 4

Q.47 Read the passage carefully and answer the Questions

Carbonyl compounds such as aldehydes, ketones, and carboxylic acids play an important role in the biochemical process of plants and animal kingdom. Many aldehydes and ketones find applications in dye, perfumery and other industries.

The carbonyl carbon atom is sp^2 -hybridized forming three sigma and one pi-bond with oxygen.

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Primary and secondary alcohols yield aldehydes and ketones respectively upon oxidation with variety of oxidizing agents.

The boiling points of aldehydes and ketones differ from hydrocarbons and ethers.

Aldehydes and ketones undergo nucleophilic addition reaction due to presence of electrophilic carbonyl carbon.

Nucleophile and a proton add across the carbon oxygen double bond.

The alcohol which cannot be used to prepare carbonyl compound through oxidation is

1. $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$
2. $(\text{CH}_3)_3\text{C-OH}$
3. $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$
4. $\text{CH}_3\text{CH}=\text{CHCH}_2\text{OH}$

Options

1. 1
2. 2
3. 3
4. 4

Q.48 Read the passage carefully and answer the Questions

Carbonyl compounds such as aldehydes, ketones, and carboxylic acids play an important role in the biochemical process of plants and animal kingdom. Many aldehydes and ketones find applications in dye, perfumery and other industries.

The carbonyl carbon atom is sp^2 -hybridized forming three sigma and one pi-bond with oxygen.

C-O bonds are polar bond due to high electronegativity of oxygen than carbon.

Hence the carbonyl carbon behaves as an electrophilic center and oxygen as a nucleophilic centre.

Primary and secondary alcohols yield aldehydes and ketones respectively upon oxidation with variety of oxidizing agents.

The boiling points of aldehydes and ketones differ from hydrocarbons and ethers.

Aldehydes and ketones undergo nucleophilic addition reaction due to presence of electrophilic carbonyl carbon.

Nucleophile and a proton add across the carbon oxygen double bond.

Identify the product Y in the following reaction



1. $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$
2. $\text{CH}_3\text{CH}=\text{CHCHO}$
3. $\text{CH}_3\text{CH}=\text{CHCOOH}$
4. $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$

Options

1. 1
2. 2
3. 3
4. 4

Q.49 Read the passage carefully and answer the Questions

Carbonyl compounds such as aldehydes, ketones, and carboxylic acids play an important role in the biochemical process of plants and animal kingdom. Many aldehydes and ketones find applications in dye, perfumery and other industries.

The carbonyl carbon atom is sp^2 -hybridized forming three sigma and one pi-bond with oxygen.

C-O bonds are polar bond due to high electronegativity of oxygen than carbon.

Hence the carbonyl carbon behaves as an electrophilic center and oxygen as a nucleophilic centre.

Primary and secondary alcohols yield aldehydes and ketones respectively upon oxidation with variety of oxidizing agents.

The boiling points of aldehydes and ketones differ from hydrocarbons and ethers.

Aldehydes and ketones undergo nucleophilic addition reaction due to presence of electrophilic carbonyl carbon.

Nucleophile and a proton add across the carbon oxygen double bond.

In polar C-O bond in carbonyl compounds, the nature of carbon and oxygen atom is

1. The carbon atom acts as a Lewis acid and the oxygen atom acts as Lewis base.
2. Both carbon and oxygen atoms act as Lewis acid
3. Both carbon and oxygen atoms act as Lewis base
4. The carbon atom acts as an Lewis base and the oxygen atom acts as a Lewis acid

Options

1. 1
2. 2
3. 3
4. 4

Q.50 Read the passage carefully and answer the Questions

Carbonyl compounds such as aldehydes, ketones, and carboxylic acids play an important role in the biochemical process of plants and animal kingdom. Many aldehydes and ketones find applications in dye, perfumery and other industries.

The carbonyl carbon atom is sp^2 -hybridized forming three sigma and one pi-bond with oxygen.

C-O bonds are polar bond due to high electronegativity of oxygen than carbon.

Hence the carbonyl carbon behaves as an electrophilic center and oxygen as a nucleophilic centre.

Primary and secondary alcohols yield aldehydes and ketones respectively upon oxidation with variety of oxidizing agents.

The boiling points of aldehydes and ketones differ from hydrocarbons and ethers.

Aldehydes and ketones undergo nucleophilic addition reaction due to presence of electrophilic carbonyl carbon.

Nucleophile and a proton add across the carbon oxygen double bond.

Formalin which is used to preserve biological specimen is an aqueous solution of

1. 40% Formaldehyde
2. 50% Formaldehyde
3. 20% Formaldehyde
4. 10% Formaldehyde

Options

1. 1
2. 2
3. 3
4. 4